When More Text Is Not Better: A Leakage-Aware Study of Text Representations for Sequential Recommendation

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Abstract

Text features are widely used to enhance sequential recommenders, yet most evidence comes from leave-one-out (LOO) protocols that allow training and test interactions to overlap in time. We conduct a leakage-aware controlled study of four text representations—ID-only, lexical TF-IDF, single-view large language model (LLM) embeddings, and prompt-based multi-view LLM features—under global temporal split (GTS) and clean LOO on Amazon Beauty and Toys. Our primary GTS protocol uses one target per user (**GTS-Last**; we also report **GTS-Random**), warm targets in the train catalog, and train-cutoff–fit text features. Three findings emerge. First, the protocol changes conclusions: under clean LOO, LLM/multi-view look strongest with inflated absolutes, whereas under GTS-Last (warm) the ranking becomes Beauty MV≈LLM≳TF and Toys MV≈UniSRec≈TF>LLM, with gaps near seed variance—so relative order among text variants is not robust. Second, across Main and Early temporal windows, TF-IDF remains competitive and often most stable, while LLM/multi-view do not exceed it by one standard deviation. Third, multi-view does not yield reliable gains: pairwise CKA among four prompt views is 0.82–0.94, concatenating views collapses effective rank to ≈53% of dimensionality, and leave-one-view-out changes MRR by at most 0.03 percentage points. In contrast, CKA between TF-IDF and LLM views is only ≈0.19–0.20, suggesting lexical and semantic features are complementary rather than redundant. Together, these results argue that strict temporal evaluation can reverse rankings among text representations, and that more semantic views do not automatically provide more useful information.

CCS Concepts

• **Information systems** → **Recommender systems**; *Content ranking*; Retrieval efficiency and effectiveness.

Keywords

Sequential Recommendation, Text Features, Temporal Evaluation, Multi-View Representation, Leakage-Aware Protocol

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1 Introduction

Sequential recommendation predicts a user's next item from historical interactions [3, 7, 16]. Item text—titles, descriptions, and LLM-encoded semantics—has become a standard way to enrich ID embeddings [1, 4, 5]. A common intuition is that richer text should help: LLM embeddings should beat bag-of-words, and multiple prompt views should beat a single view.

This intuition is fragile under realistic evaluation. Most text-enhanced sequential recommenders are still validated with leave-one-out (LOO) splits, which can leak future interactions into training and inflate absolute metrics [6, 15]. Global temporal split (GTS) enforces a train/valid/test cutoff and better matches deployment, but it also exposes distribution shift: catalog turnover, cold targets, and popularity decay. Prior work has studied split choices in general [9, 11]; our focus is narrower and more actionable for practitioners who choose among *text representations*: does the protocol change the ranking of TF-IDF vs. single-view LLM vs. prompt multi-view?

We study this question with a controlled, leakage-aware protocol. All text features (vocabulary/TF-IDF, SVD, centering/whitening, LLM embeddings) are fit only on items observed before the training cutoff. We compare five systems on Amazon Beauty and Toys under clean LOO and GTS, with a frozen Early temporal window and UniSRec [5] as an external text baseline. We further diagnose multi-view failure with CKA [8], effective rank, and leave-one-view-out (LOOV).

Research questions.

RQ1 Does switching from LOO to GTS change conclusions about TF-IDF, single-view LLM, and multi-view text?

RQ2 Which text representation is most robust across temporal windows under distribution shift?

RQ3 Why does prompt-based multi-view fail to reliably beat single-view?

Contributions. (1) A leakage-aware comparison showing that GTS reverses LOO rankings among text representations. (2) Evidence that TF-IDF is the most stable under Main/Early windows, associated with (not claimed as caused by) rising unseen-target share and catalog turnover. (3) Triangulated redundancy analysis (CKA + effective rank + LOOV) explaining why more prompt views do not help, while lexical TF-IDF remains complementary to semantic LLM features.

2 Related Work

Text-enhanced sequential recommendation. Early work fused bag-of-words or review text with collaborative filtering [14, 18]. Recent systems encode titles/descriptions with pretrained language models and transfer them across domains [1, 2, 4, 5]. We deliberately do *not* propose a new architecture; instead we ask whether stronger text encoders and more views remain advantageous once temporal leakage is controlled.

Evaluation protocols and temporal leakage. Sampled metrics can distort rankings [9]. LOO vs. time-based splits systematically change

absolute scores and can alter model selection [6, 11, 15]. We follow this line but specialize it to text representations: protocol effects matter precisely because TF-IDF, LLM, and multi-view respond differently to temporal shift.

Multi-view learning. Multi-view theory emphasizes *useful complementary* information rather than "more views" [19]. Representation similarity tools such as CKA [8] help detect redundancy. Our prompt views share the same encoder and item title, making redundancy a first-order risk.

3 Leakage-Aware Protocol and Models

3.1 Temporal Protocols

Global Temporal Split (GTS).. Interactions are ordered by timestamp. We use RecBole's TS split [20] with ratios [0.8, 0.1, 0.1] (Main) or [0.6, 0.1, 0.3] (Early). A global cutoff places the final 10% (Main) of interactions into test; train/valid are further split within the pre-test window. Training still uses successive next-item steps (standard SASRec augmentation). For *evaluation*, All-target GTS (Successive-All) mismatches standard next-item prediction, so our **primary** metrics use:

- **GTS-Last**: one last test target per user;
- **GTS-Random**: one uniformly sampled test target per user (fixed per-user RNG seed).

Warm metrics further restrict targets to the train catalog. Successive-All numbers are reported only in Appendix A as a RecBole-native reference.

Clean LOO.. For contrast, we also run leave-one-out evaluation without temporal cutoff (train/valid/test as last-3 interactions per user), using the same model family and hyperparameters. LOO is *not* used for main conclusions.

Leakage controls. TF-IDF vocabulary/IDF, SVD projection, and LLM center/whitening are fit **only** on items appearing before the training cutoff. Evaluation optionally restricts to users with ≥ 5 training interactions (`min_train_interactions=5`). Warm-target analyses keep only targets present in the train catalog; unseen targets are reported separately (Appendix C).

3.2 Model Variants

All models share a SASRec [7] backbone (d=64, 2 blocks) unless noted.

- **ID**: ID embeddings only.
- **TF**: train-cutoff TF-IDF (SVD to 256-d) fused via a lightweight text head + InfoNCE alignment [17].
- **LLM**: single-view Qwen2.5-7B [13] item embedding (TS-whitened), same fusion.
- **MV**: four prompt views (Description / Function / Audience / Style), concatenated then projected; no cross-attention in the main table.
- **UniSRec**: external text sequential baseline [5] with the same TS embeddings and GTS filter for fair comparison.

Two-phase training (text-head warmup, then joint fine-tuning) is used for TF/LLM/MV. Cross-attention variants are relegated to Appendix D as negative results.

Table 1: Primary main table: GTS-Last MRR@10 (3-seed mean±std, `min_train=5`). Warm = target in train catalog (recommended for main claims). Bold: best among ID/TF/LLM/MV.

Dataset	Scope	ID	TF	LLM	MV	UniSRec
Beauty	All-user	.0022±.0004	.0036±.0006	.0041±.0003	**.0042±.0004**	.0036±.0001
	Warm	.0032±.0006	.0051±.0008	.0059±.0005	**.0061±.0005**	.0052±.0002
Toys	All-user	.0019±.0001	.0025±.0003	.0022±.0001	**.0026±.0004**	.0026±.0000
	Warm	.0031±.0002	.0040±.0005	.0035±.0002	**.0042±.0006**	.0041±.0001

Table 2: Protocol contrast (MRR@10, 3-seed mean±std). LOO vs. primary GTS-Last (all-user) vs. GTS-All reference. LOO inflates absolutes and favors LLM/MV; GTS-Last reorders close text variants.

Dataset	Protocol	ID	TF	LLM	MV
Beauty	LOO	.0175±.0004	.0251±.0002	**.0254±.0004**	.0253±.0004
	GTS-Last	.0022±.0004	.0036±.0006	.0041±.0003	**.0042±.0004**
	GTS-All†	.0106±.0005	**.0165±.0008**	.0161±.0005	.0160±.0008
Toys	LOO	.0152±.0005	.0247±.0006	**.0258±.0012**	**.0258±.0028**
	GTS-Last	.0019±.0001	.0025±.0003	.0022±.0001	**.0026±.0004**
	GTS-All†	.0070±.0001	**.0113±.0003**	.0107±.0004	.0109±.0006

† Appendix reference (Successive-All); not used for main claims.

4 Experimental Setup

Datasets. Amazon Beauty and Toys & Games [10, 12]. Beauty: 1.21M users, 259K items, 2.02M interactions. Toys: larger catalog (336K items) with similar sparsity. After GTS + `min_train=5` filtering, Beauty has 4,520 test users (12,001 All-targets) and Toys 4,873 users (13,186 All-targets).

Metrics. Full-ranking Hit@10, NDCG@10, MRR@10 [9]. Main tables report MRR@10 (mean±std over seeds {42, 2024, 2026}).

Compute. All runs use RecBole [20] with identical full-sort evaluation. Hyperparameters for Early are frozen from Main (no re-tuning). Parameter/latency costs are nearly identical across ID/TF/LLM/MV (Table 6).

5 Results

5.1 RQ1: Protocol Changes Rankings

Table 1 is our primary result under the locked protocol (GTS-Last, `min_train=5`). Text still beats ID, but **no text variant dominates by more than about one std**: Beauty warm ranks MV≈LLM≳TF≈UniSRec; Toys warm ranks MV≈UniSRec≈TF>LLM. Table 2 answers RQ1: clean LOO inflates MRR by an order of magnitude vs. Last and favors LLM/MV, while Successive-All (appendix reference) instead slightly favors TF-IDF. Thus the protocol does not merely rescale scores—it changes which text representation looks best. This specializes prior temporal-evaluation warnings [6, 15] to TF/LLM/MV. GTS-Random (same checkpoints, one random test target per user) is reported alongside Last once computed; we treat agreement between Last and Random as a robustness check for the primary ranking story.

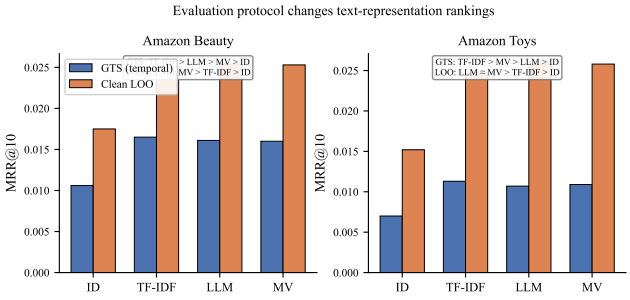


Figure 1: LOO vs. GTS-All contrast (reference view). Primary claims use GTS-Last/Random in Table 1; All is shown here because it matches RecBole successive evaluation and makes the LOO inflation visually clear.

Table 3: Main vs. Early temporal window on Beauty (MRR@10 test, 3 seeds, GTS-All). Hyperparameters frozen from Main. Early: TS[0.6, 0.1, 0.3].

Window	TF-IDF	LLM	MV
Main (GTS-All)	**.0165**±.0008	.0161±.0005	.0160±.0008
Early60	**.0053**±.0002	.0049±.0002	.0052±.0001

Table 4: Temporal distribution shift (valid→test under Main GTS). We report *associations*, not causal claims.

Dataset	Test new-item	Test unseen bucket	Pop mean (V→T)	Pop p50 (V→T)
Beauty	27.3%	17.8%	112.8 → 101.8	21 → 15
Toys	32.8%	21.3%	144.7 → 119.1	18 → 11

External baseline. Under the same GTS-Last filter, UniSRec is 0.0052±.0002 (Beauty warm) and 0.0041±.0001 (Toys warm)—competitive with TF but not above MV/LLM on Beauty. Equalizing the eval filter removes earlier apparent UniSRec wins under mismatched settings.

5.2 RQ2: Temporal Robustness

Table 3 shows that Early window preserves the Main ordering: TF-IDF remains strongest; LLM/MV never exceed it by one std. Absolute scores drop sharply (Main→Early), concurrent with the distribution statistics in Table 4: higher catalog novelty, more unseen targets, and lower target popularity from valid to test. Warm-target audit (eligible users, hist≥5) similarly finds test unseen shares of 27.8% (Beauty) and 34.5% (Toys). We phrase these as associations with the valid→test performance drop (43–55% in our runs), not as proven causes.

5.3 RQ3: Why Multi-View Does Not Help

Figure 2 and Table 5 address RQ3. Four prompt views are structurally redundant: inter-view CKA ranges 0.82–0.94 (Toys similar; 3-subsample std < 0.002), while concat effective-rank ratio is only 0.53 (Beauty) / 0.55 (Toys). LOOV deltas are tiny (|Δ|<0.0004), i.e., <3% relative to full MV MRR. CKA alone does not prove uselessness [8], but CKA + effective rank + LOOV together support the claim that prompt views largely restate the same semantics. Meanwhile TF–LLM CKA ≈ 0.19–0.20 indicates lexical and semantic features are complementary—yet simply stacking four LLM prompts does not harvest that complementarity. Table 6 shows the failure is not due to capacity: MV has essentially the same parameter/latency budget as TF/LLM.

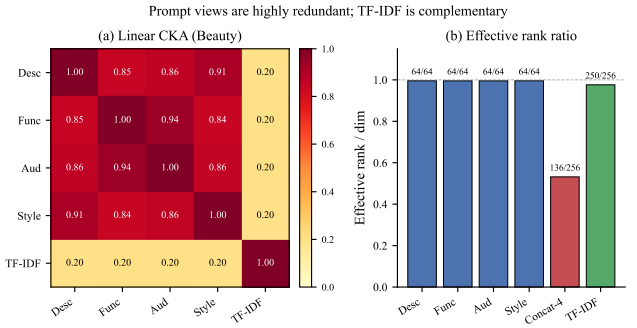


Figure 2: Representation analysis on Beauty (3 item subsamples of 3,000; mean CKA). (a) Inter-view CKA is high (0.84–0.94); TF–LLM CKA is low (≈ 0.20). (b) Concatenating four 64-d views yields effective rank 136/256 (ratio 0.53), vs. TF-IDF 250/256 (0.98).

Table 5: Leave-one-view-out (zero-mask) on GTS MV checkpoints (seed 42). Δ is MRR@10 change vs. full 4-view model.

Dataset	Full	− Desc	− Func	− Aud	− Style
Beauty	.01591	−.00006	+.00011	−.00016	−.00034
Toys	.01018	−.00019	−.00011	−.00012	−.00015

Table 6: Parameter count and inference cost (seed 42, Beauty/Toys). MV adds <1% parameters vs. ID; latency is negligible.

Model	Beauty params (M)	Toys params (M)	Beauty ms/samp.	Toys ms/samp.
ID	67.16	86.84	0.02	0.01
TF	67.43	87.10	0.02	0.02
LLM	67.56	87.24	0.02	0.02
MV	67.71	87.38	0.02	0.03
UniSRec	67.69	87.37	0.04	0.05

5.4 Takeaways for Practitioners

(1) Prefer GTS-Last/Random (warm, cutoff-fit features) when comparing text representations; LOO can reverse rankings and inflate scores.

(2) Do not treat Successive-All as the default main table; All vs. Last/Random rescales metrics and can reorder close text variants.

(3) Start with cutoff-fit TF-IDF before investing in multi-prompt LLM views.
(4) If using LLM text, a single view often matches four; diagnose redundancy before scaling views.

6 Conclusion

We reframed text-enhanced sequential recommendation as a leakage-aware empirical study rather than a race for a new SOTA model. Under the locked GTS-Last protocol, text beats ID but TF/LLM/MV gaps stay near seed variance; under LOO, LLM/multi-view look stronger with inflated absolutes; under Successive-All, TF-IDF is slightly preferred. Target selection is therefore part of the scientific claim, not a footnote. Temporal windows and distribution-shift statistics reinforce that semantic multi-view gains are not robust, and representation analyses explain why: prompt views are redundant, while lexical and semantic features are complementary but not exploited by naive multi-prompt stacking. Future work includes full Successive evaluation at scale and hybrid TF+LLM fusion that targets complementarity rather than view count.

Ethical Considerations

This work uses public Amazon review datasets without personally identifiable private attributes beyond standard user/item IDs. Potential harms include reinforcing popularity bias under temporal shift and over-trusting LLM text features that may encode stereotypes present in product descriptions. We mitigate the first by reporting stratified/warm-target diagnostics; the second by treating multi-view LLM features as an empirical object of study rather than a default upgrade. No human subjects were involved.

References

[1] Keqin Bao, Jizhi Zhang, Yang Zhang, Wenjie Wang, Fuli Feng, and Xiangnan He. 2023. TALLRec: An Effective and Efficient Tuning Framework to Align Large Language Model with Recommendation. In *Proceedings of the 17th ACM Conference on Recommender Systems (RecSys)*. Singapore, 100–110. doi:10.1145/3604915.3608834

[2] Shijie Geng, Shuchang Liu, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2022. Recommendation as Language Processing (RLP): A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5). In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*. Seattle, WA, USA, 299–315. doi:10.1145/3523227.3546738

[3] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. 2016. Session-based Recommendations with Recurrent Neural Networks. In *Proceedings of the 4th International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1511.06939

[4] Yupeng Hou, Zhankui He, Julian McAuley, and Wayne Xin Zhao. 2023. Learning Vector-Quantized Item Representation for Transferable Sequential Recommenders. In *Proceedings of the ACM Web Conference 2023 (WWW)*. Austin, TX, USA, 2372–2383. doi:10.1145/3543507.3583315

[5] Yupeng Hou, Shanlei Mu, Wayne Xin Zhao, Yaliang Li, Bolin Ding, and Ji-Rong Wen. 2022. Towards Universal Sequence Representation Learning for Recommender Systems. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. Washington, DC, USA, 585–593. doi:10.1145/3534678.3539053

[6] Yitong Ji, Aixin Sun, and Jie Zhang. 2023. Towards Realistic Evaluation of Recommender Systems. In *Proceedings of the 17th ACM Conference on Recommender Systems (RecSys)*. Discusses temporal leakage and leave-one-out vs. global temporal split.

[7] Wang-Cheng Kang and Julian McAuley. 2018. Self-Attentive Sequential Recommendation. In *Proceedings of the 2018 IEEE International Conference on Data Mining (ICDM)*. 197–206. doi:10.1109/ICDM.2018.00035

[8] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of Neural Network Representations Revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*. 3519–3529.

[9] Walid Krichene and Steffen Rendle. 2020. On Sampled Metrics for Item Recommendation. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*. 1618–1627. doi:10.1145/3394486.3403226

[10] Julian McAuley, Christopher Targett, Qinfeng Shi, and Anton van den Hengel. 2015. Image-based Recommendations on Styles and Substitutes. In *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 43–52. doi:10.1145/2766462.2767755

[11] Zaiqiao Meng, Richard McCreadie, Craig Macdonald, and Iadh Ounis. 2020. Temporal Collaborative Filtering with Graph Convolutional Networks. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. 1789–1792. doi:10.1145/3397271.3401321

[12] Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying Recommendations using Distantly-Labeled Reviews and Fine-Grained Aspects. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 188–197. doi:10.18653/v1/D19-1018

[13] Qwen Team. 2024. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115* (Dec. 2024). doi:10.48550/arXiv.2412.15115 Alibaba Cloud. The report accompanies open-weight releases and documents hosted MoE endpoints (Qwen2.5-Turbo, Qwen2.5-Plus) in Alibaba Cloud Model Studio..

[14] Gerard Salton and Christopher Buckley. 1988. Term-Weighting Approaches in Automatic Text Retrieval. *Information Processing & Management* 24, 5 (1988), 513–523. doi:10.1016/0306-4573(88)90021-0

[15] Aixin Sun and Yitong Ji. 2023. Revisiting Evaluation Protocols for Sequential Recommendation. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. Compares LOO and time-based splits for next-item recommendation.

[16] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*. 1443–1452. doi:10.1145/3357384.3357895

[17] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation Learning with Contrastive Predictive Coding. *arXiv preprint arXiv:1807.03748* (2018). doi:10.48550/arXiv.1807.03748

[18] Hao Wang, Naiyan Wang, and Dit-Yan Yeung. 2015. Collaborative Deep Learning for Recommender Systems. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 1235–1244. doi:10.1145/2783258.2783273

[19] Chang Xu, Dacheng Tao, and Chao Xu. 2013. A Survey on Multi-view Learning. *arXiv preprint arXiv:1304.5634* (2013). Survey emphasizing complementary information across views.

[20] Wayne Xin Zhao, Shanlei Mu, Yupeng Hou, Zihan Lin, Yushuo Chen, Xingyu Pan, Kaiyuan Li, Yujie Lu, Hui Wang, Changxin Tian, Yingqian Min, Zhichao Feng, Xinyan Fan, Xu Chen, Pengfei Wang, Wendi Ji, Yaliang Li, Xiaoling Wang, and Ji-Rong Wen. 2021. RecBole: Towards a Unified, Comprehensive and Efficient Framework for Recommendation Algorithms. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*. 4653–4664. doi:10.1145/3459637.3482010

A GTS-All Reference (Successive-All)

RecBole's native post-cutoff evaluation scores *all* successive test targets per user (Beauty: 4,520 users → 12,001 samples; Toys: 4,873 → 13,186). Under this All protocol (MRR@10, `min_train=5`): Beauty TF 0.0165±.0008, LLM 0.0161, MV 0.0160, UniSRec 0.0123; Toys TF 0.0113±.0003, MV 0.0109, LLM 0.0107, UniSRec 0.0082. We keep All only as a reference because it mismatches standard next-item prediction; main claims use Last/Random (Table 1).

B GTS-Random

GTS-Random draws one test target per user with a fixed per-user RNG seed (base seed 42). Results are produced with `scripts/eval_gts_random.py` on the same checkpoints as Table 1 and will be merged into the main Last/Random table upon completion (`paper_recsys/gts_random_eval.csv`).

C Warm / Unseen Notes

Primary warm metrics are already in Table 1 (GTS-Last). Unseen-target shares under eligible users (hist≥5) are about 27.8% (Beauty)

and 34.5% (Toys); models lack a content-only zero-shot fallback, so unseen results stay diagnostic.

D Cross-Attention (Negative Result)

Optional sequence–text cross attention improves and degrades roughly equally across users; score entropy/margin shifts are unstable across seeds. We therefore exclude Cross from the main method and title, retaining it only as a negative appendix observation.